

SECTION A- MCQ (20 Marks: 1Mks /Correct Answer)

1. What is the primary goal of Artificial Intelligence?
 - a) To replace human intelligence
 - b) To create machines that can perform tasks requiring human intelligence
 - c) To develop only theoretical models
 - d) To focus solely on robotics
2. Which of the following is NOT a type of AI?
 - a) Narrow AI
 - b) General AI
 - c) Super AI
 - d) Reactive AI
3. What is the primary purpose of Machine Learning?
 - a) To program explicit rules for decision-making
 - b) To enable systems to learn from data and improve over time
 - c) To replace human decision-making entirely
 - d) To focus only on data storage
4. Which of the following is a supervised learning algorithm?
 - a) K-Means Clustering
 - b) Decision Trees
 - c) Principal Component Analysis (PCA)
 - d) Apriori Algorithm
5. What is the role of a cost function in Machine Learning?
 - a) To evaluate the performance of a model
 - b) To preprocess the data
 - c) To visualize the data
 - d) To store the dataset
6. Which of the following is an example of unsupervised learning?
 - a) Linear Regression
 - b) Logistic Regression
 - c) K-Means Clustering
 - d) Support Vector Machines (SVM)
7. What is overfitting in Machine Learning?
 - a) When a model performs well on training data but poorly on unseen data
 - b) When a model is too simple to capture patterns in the data
 - c) When a model fails to converge during training
 - d) When a model is trained on insufficient data
8. What is the main ethical concern regarding AI and ML?
 - a) High computational cost
 - b) Bias in algorithms
 - c) Lack of data availability
 - d) Complexity of models
9. Which of the following is an example of reinforcement learning?
 - a) Training a robot to navigate a maze
 - b) Classifying emails as spam or not spam
 - c) Grouping customers based on purchasing behavior
 - d) Predicting house prices
10. What is the purpose of cross-validation in ML?
 - a) To reduce the size of the dataset
 - b) To evaluate the model's performance on unseen data
 - c) To preprocess the data
 - d) To visualize the data
11. What is the role of the learning rate in gradient descent?
 - a) It determines the number of iterations required for convergence
 - b) It controls the step size during parameter updates
 - c) It defines the number of layers in a neural network
 - d) It measures the model's accuracy
12. Which of the following is a hyperparameter in a neural network?
 - a) Weights

- b) Bias
- c) Learning rate
- d) Activation function

13. What is the output of a logistic regression model?

- a) A continuous value
- b) A probability value between 0 and 1
- c) A binary value (0 or 1)
- d) A categorical label

14. What is the output of a linear regression model?

- a) A continuous value
- b) A probability value between 0 and 1
- c) A binary value (0 or 1)
- d) A categorical label

15. What is the purpose of the confusion matrix?

- a) To evaluate the performance of a regression model
- b) To visualize the performance of a classification model
- c) To reduce the dimensionality of the data
- d) To optimize hyperparameters

16. Which of the following is a loss function used in linear regression?

- a) Cross-Entropy Loss
- b) Hinge Loss

- c) Mean Squared Error (MSE)
- d) Log Loss

17. Which of the following is NOT a clustering algorithm?

- a) K-Means
- b) DBSCAN
- c) Hierarchical Clustering
- d) Linear Regression

18. Which of the following is a popular deep learning framework?

- a) Scikit-learn
- b) TensorFlow
- c) Weka
- d) RapidMiner

19. What is the purpose of an activation function in a neural network?

- a) To normalize the input data
- b) To introduce non-linearity into the model
- c) To reduce the dimensionality of the data
- d) To store the weights of the network

20. Which of the following is an application of Natural Language Processing (NLP)?

- a) Image recognition
- b) Sentiment analysis
- c) Fraud detection
- d) Clustering

SECTION B: Question 1 Compulsory; Answer Question 2 or 3, (50 MARKS)

QUESTION 1 (10 Mks)

From the following Confusion Matrix

1. Identify TP, FP, TN, and FN (4 Mks)
2. Give the formula and calculate:
 - a. False Positive Rate (2 Mks)
 - b. True Positive Rate (2 Mks)
 - c. F1-Score (2 Mks)

	Predicted Positive	Predicted Negative
Actual Positive	80	20
Actual Negative	10	90

QUESTION 2**(40 Mks)**

Considering the *Digits* dataset for optical character recognition (OCR) from *SKLearn* library. Give the syntax to accomplish the following actions:

1. Import the various libraries (3 Mks)
2. Import the Dataset (3 Mks)
3. Split the Dataset in 85% for training (3 Mks)
4. For the **SVM** classifier
 - a. Import and initiate it (3 Mks)
 - b. Train the Model (3 Mks)
 - c. Evaluate the Score (3 Mks)
 - d. Predict the corresponding values for the Test set (3 Mks)
5. For the **Logistic Regression** classifier
 - a. Import and initiate it (3 Mks)
 - b. Train the Model (3 Mks)
 - c. Evaluate the Score (3 Mks)
 - d. Predict the corresponding values for the Test set (3 Mks)
6. Import the **CROSSVALIDATION** module and pass in a Logistic Regression Classifier (3 Mks)
7. Generate the confusion Matrix corresponding for any of these classifiers (3 Mks)
8. Plot the confusion Matrix with the Heatmap with *seaborn* (3 Mks)

QUESTION 3**(40 Mks)**

Let's Consider the IRIS flower's data set, from the SKLEARN library. We want to use Artificial Neural Network model to build a classifier

1. Sketch an ANN made up of one hidden layer of 7 Neurons to fit this dataset, mention the various activations functions as well. (10 Mks)
2. How many parameters will be computed for this network? (2 Mks)
3. With the use of TensorFlow framework,
 - a. Import the dataset (4 Mks)
 - b. Sequentially Build the network (4 Mks)
 - c. Compile it with the correct optimizer, Loss, and related Metrics (4 Mks)
 - d. Check the summary of the model (4 Mks)
 - e. Train it (4 Mks)
 - f. Check the score on the testing Set (4 Mks)
 - g. Plot the confusion Matrix (4 Mks)

Good Luck